
Gen Chem 7

thermochemistry

M-CATS @ university of kentucky
9/25 chem session 6:00 PM

-I-

Energy moves

Energy moves

System

Surroundings

First law of thermo

The 4 types of thermodynamic processes:

State Functions

Standard Conditions

Heat versus Temperature

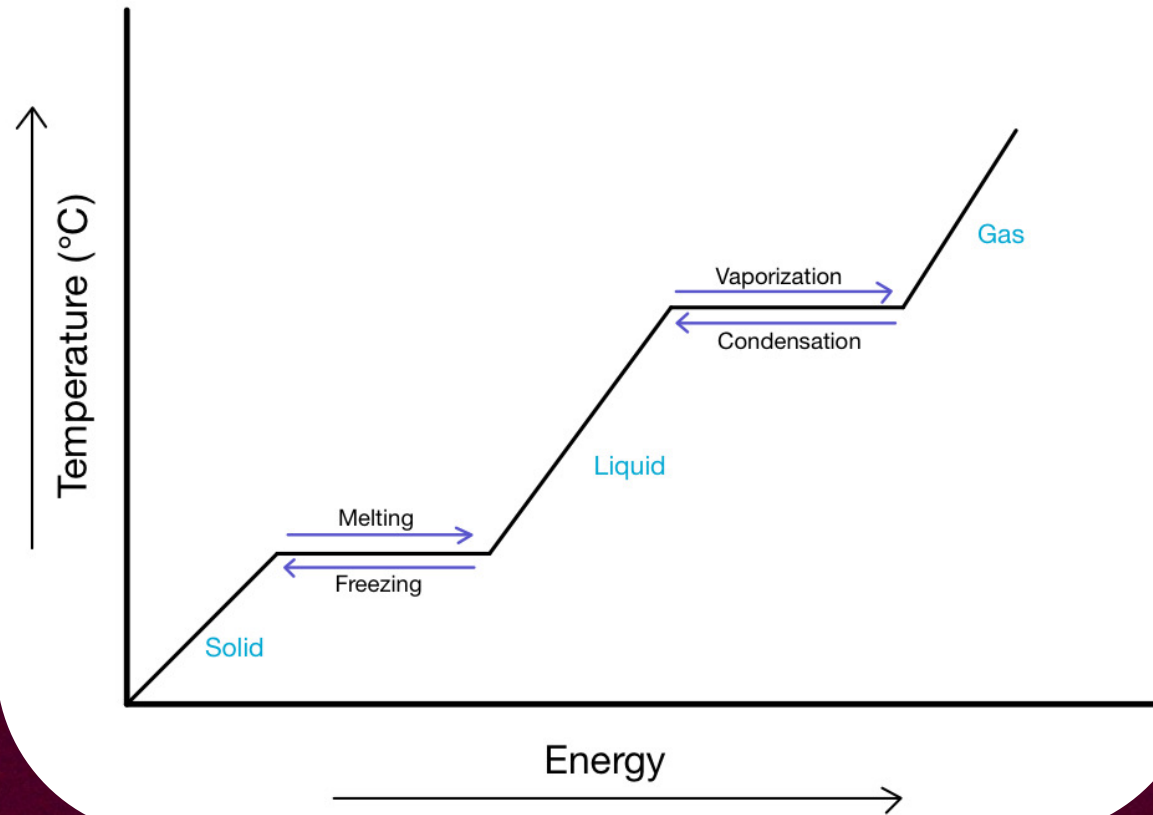
Heat

Temperature

$$q=mc\Delta T$$

Going through (phase) changes

Phase Change Diagram



$$q = mL$$

-II-

The Big 3

Enthalpy

Entropy

Gibbs Free Energy

Equilibrium

-III-

Questions

A gas is rapidly compressed in an insulated container.

- a)** Is the process adiabatic, isothermal, isobaric, or isochoric?
- b)** Is heat transferred into or out of the system?
- c)** What happens to the internal energy of the gas?

A 50 g sample of water absorbs 2.09 kJ of heat. The specific heat of water is:

$c = 4.18 \text{ J/(g} \cdot ^\circ\text{C)}$ What is the temperature change?

A. 1°C

B. 5°C

C. 10°C

D. 20°C

Consider:



What is ΔH for:



- A. $+25 \text{ kJ}$
 - B. -25 kJ
 - C. $+100 \text{ kJ}$
 - D. -100 kJ
-

A reaction has:

$\Delta H = -40 \text{ kJ/mol}$ and

$\Delta S = -100 \text{ J/(mol}\cdot\text{K)}$

Which statement is correct?

A. The reaction is spontaneous at all temperatures.

B. The reaction is never spontaneous.

C. The reaction is favored at low temperatures but not necessarily at high temperatures.

D. The reaction is favored at high temperatures but not necessarily at low temperatures.